Name:Tharunya BS Register Number:250701786

Experiment Name:Experiment to understand EDA-Quantitative and Qualitative Analysis

```
import seaborn as sns
import pandas as pd import
numpy as np
import matplotlib.pyplot as plt
%matplotlib inline tips =

sns.load_dataset('tips')

print(tips.head())

sns.displot(tips['total_bill'], kde=True) plt.show()

sns.displot(tips['total_bill'], kde=False) plt.show()

sns.jointplot(x='tip', y='total_bill', data=tips) plt.show()

sns.jointplot(x='tip', y='total_bill', data=tips, kind='reg')
plt.show()

sns.jointplot(x='tip', y='total_bill', data=tips, kind='hex')
plt.show()

sns.pairplot(tips) plt.show()
print(tips['time'].value_counts())

sns.pairplot(tips, hue='time') plt.show()

sns.pairplot(tips, hue='day') plt.show()

sns.heatmap(tips.corr(numeric_only=True), annot=True) plt.show()

sns.boxplot(y=tips['total_bill']) plt.show()

sns.boxplot(y=tips['tip']) plt.show()

sns.countplot(y=tips['day']) plt.show()
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sns.countplot(y=tips['sex']) plt.show()

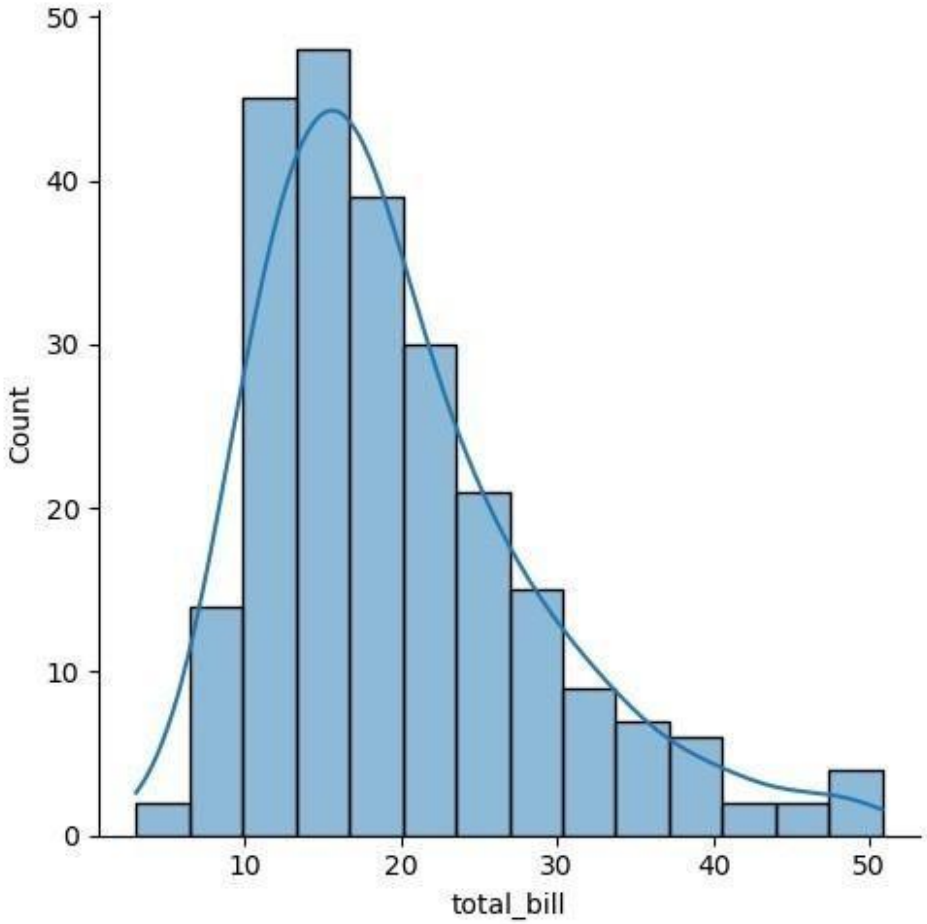
tips['sex'].value_counts().plot(kind='pie') plt.show()

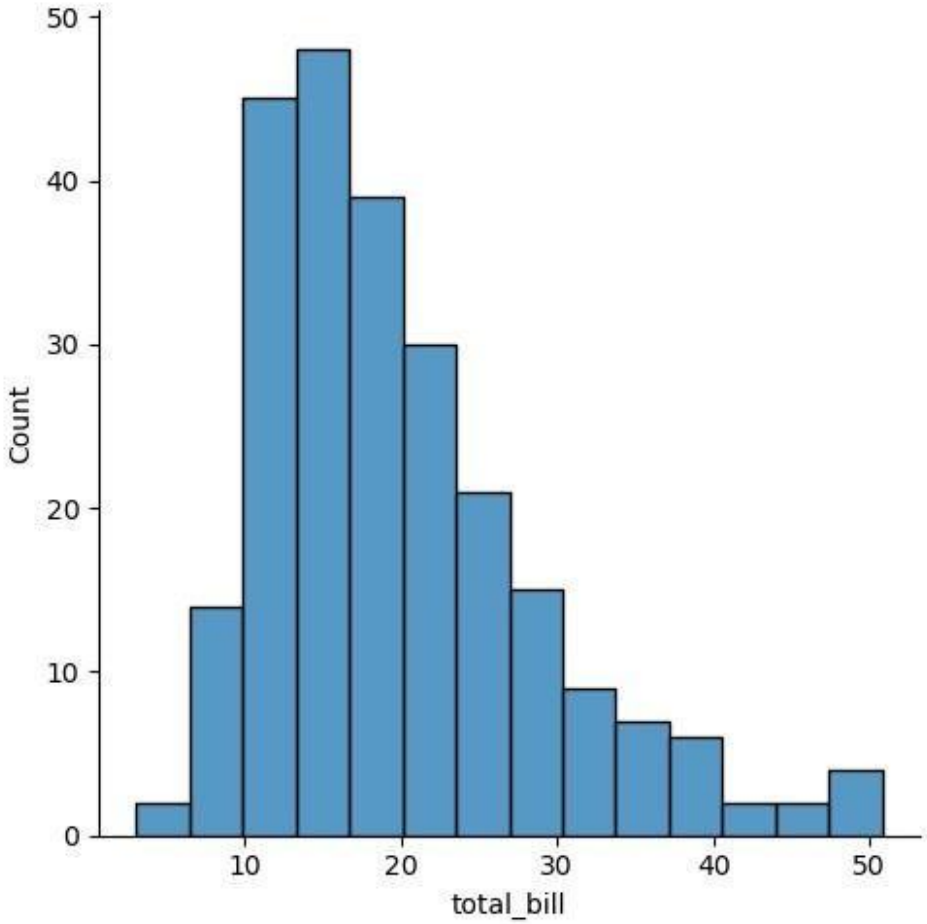
tips['sex'].value_counts().plot(kind='bar') plt.show()

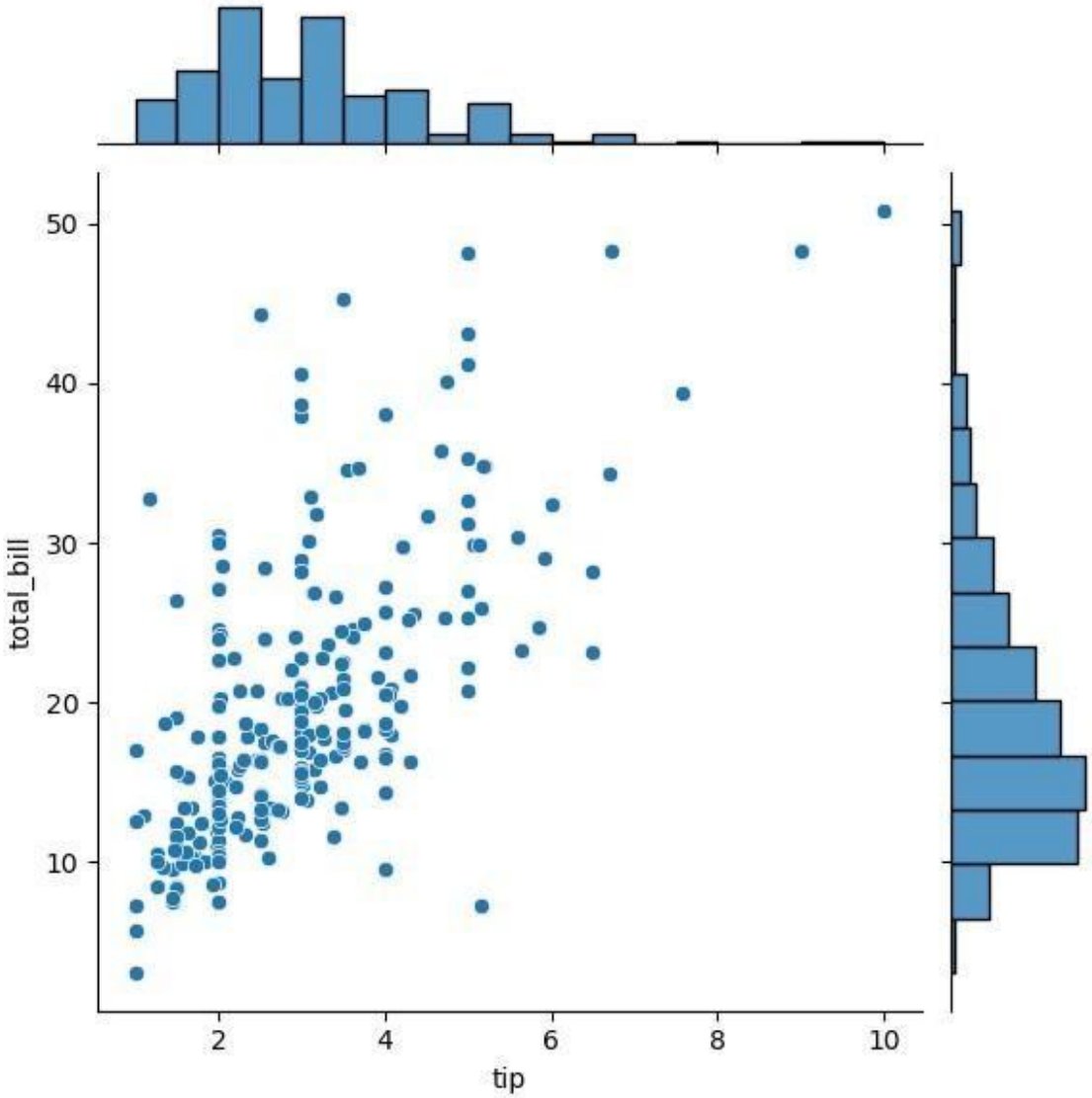
sns.countplot(x=tips['sex']) plt.show()

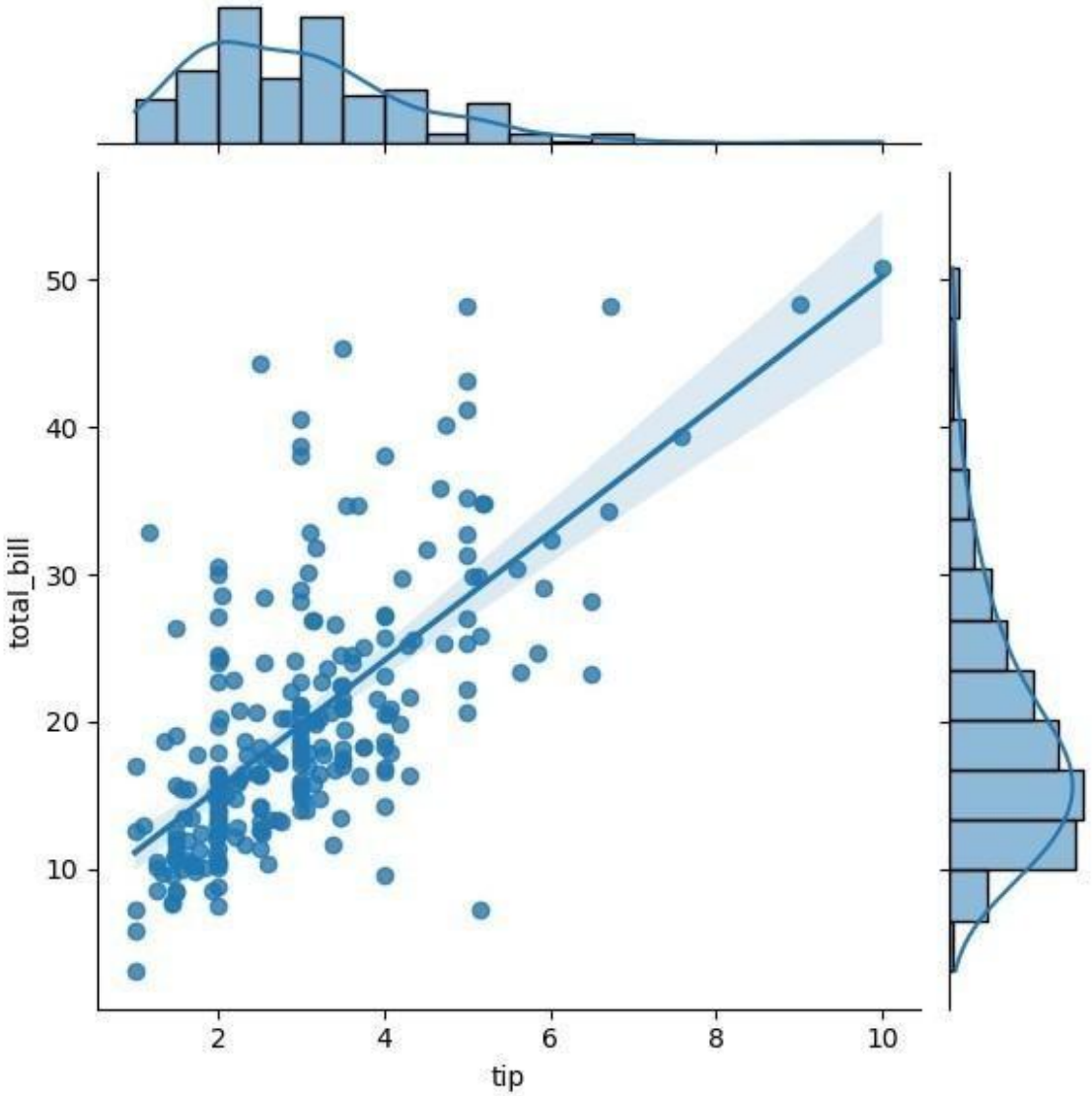
sns.countplot(data=tips[tips['time'] == 'Dinner'], x='day') plt.show()
```

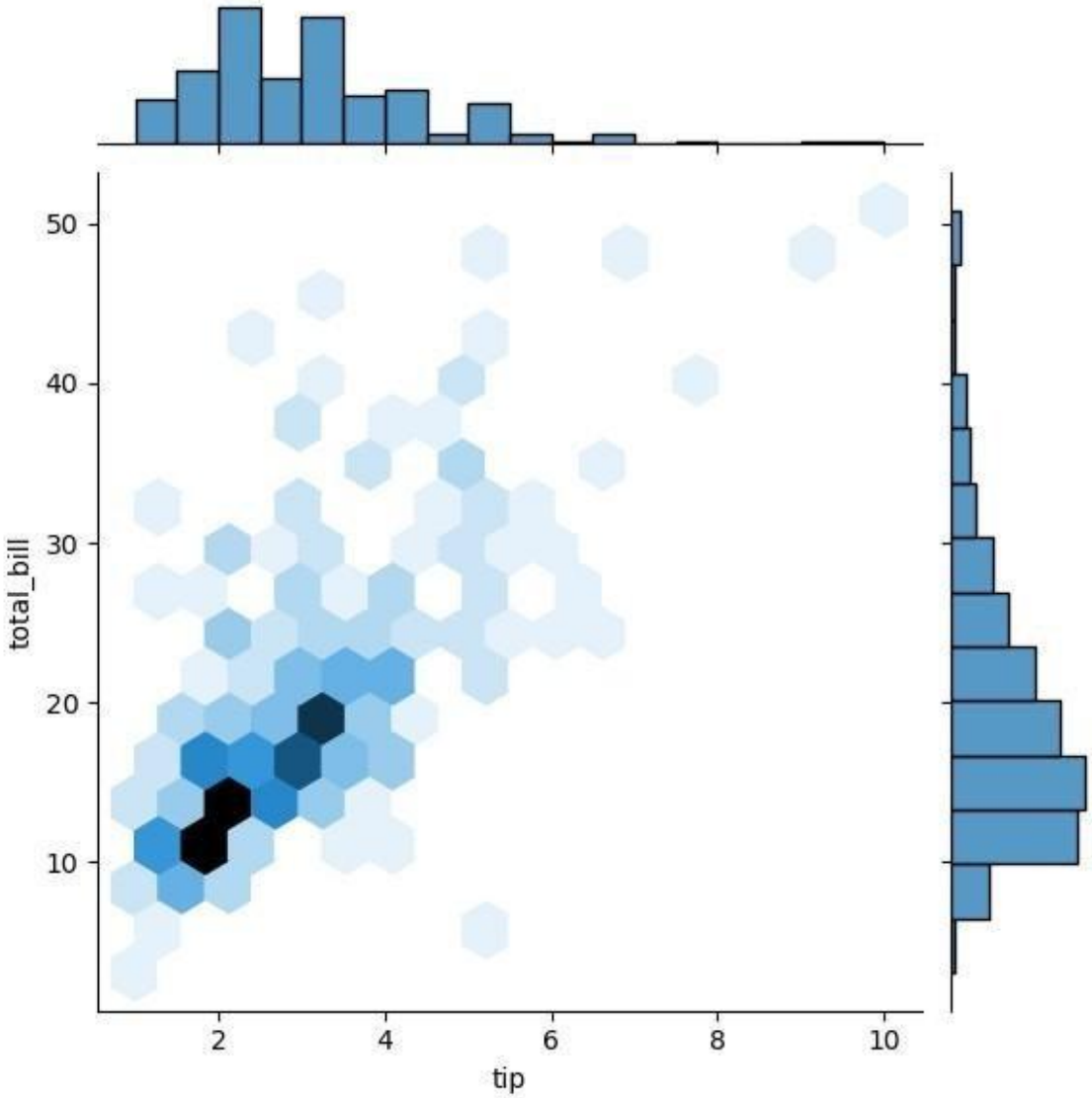
| | total_bill | tip | sex | smoker | day | time | size |
|---|---|---|---|---|---|---|---|
| 0 | 16.99 | 1.01 | Female | No | Sun | Dinner | 2 |
| 1 | 10.34 | 1.66 | Male | No | Sun | Dinner | 3 |
| 2 | 21.01 | 3.50 | Male | No | Sun | Dinner | 3 |
| 3 | 23.68 | 3.31 | Male | No | Sun | Dinner | 2 |
| 4 | 24.59 | 3.61 | Female | No | Sun | Dinner | 4 |

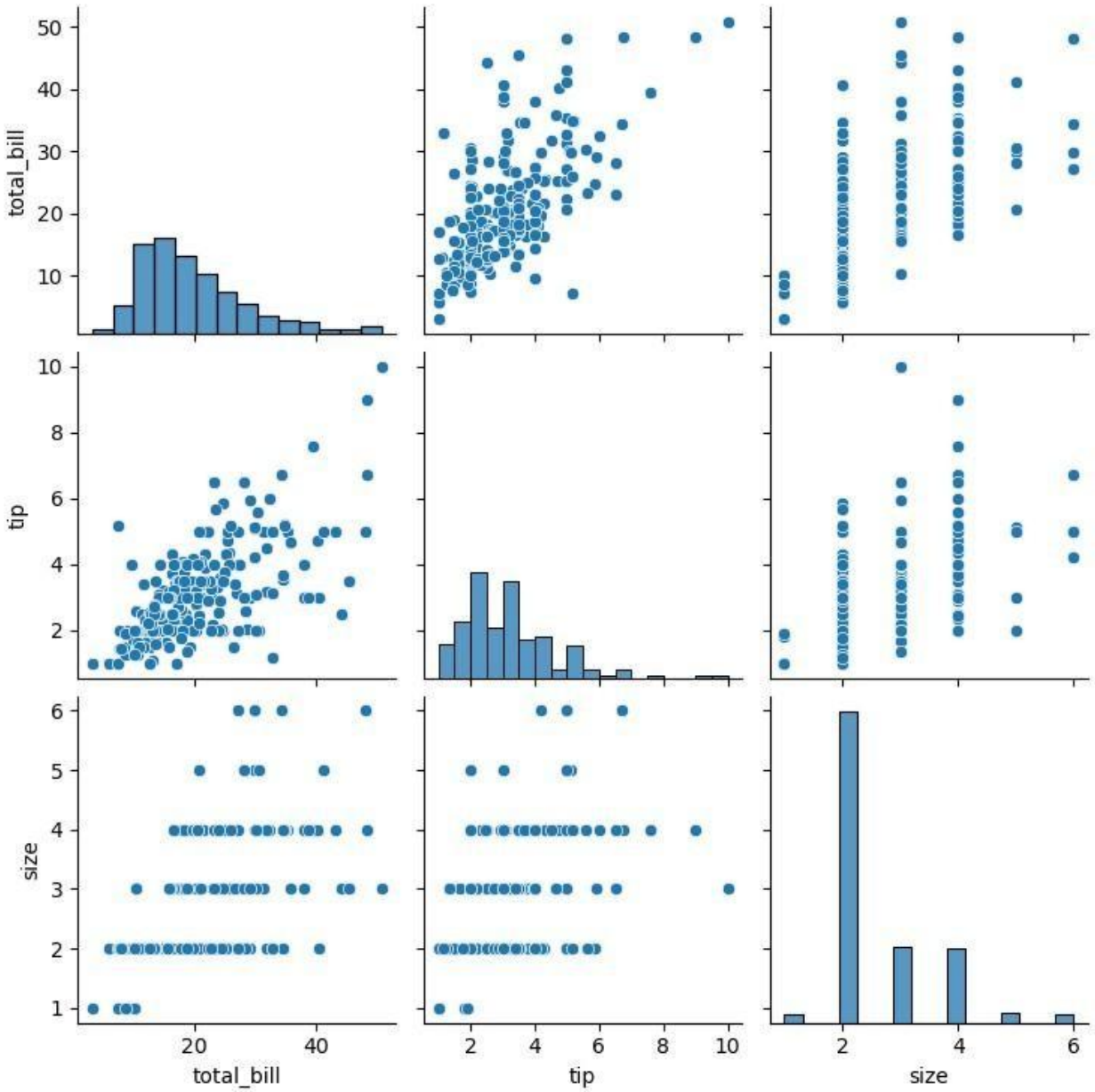

50
40
30
20
10
0
Count
10
20
30
40
50
total_bill


50
40
30
20
10
0
Count
10
20
30
40
50
total_bill


50
40
30
20
10
total_bill
2
4
6
8
10
tip


total_bill
tip
10
20
30
40
50
2
4
6
8
10

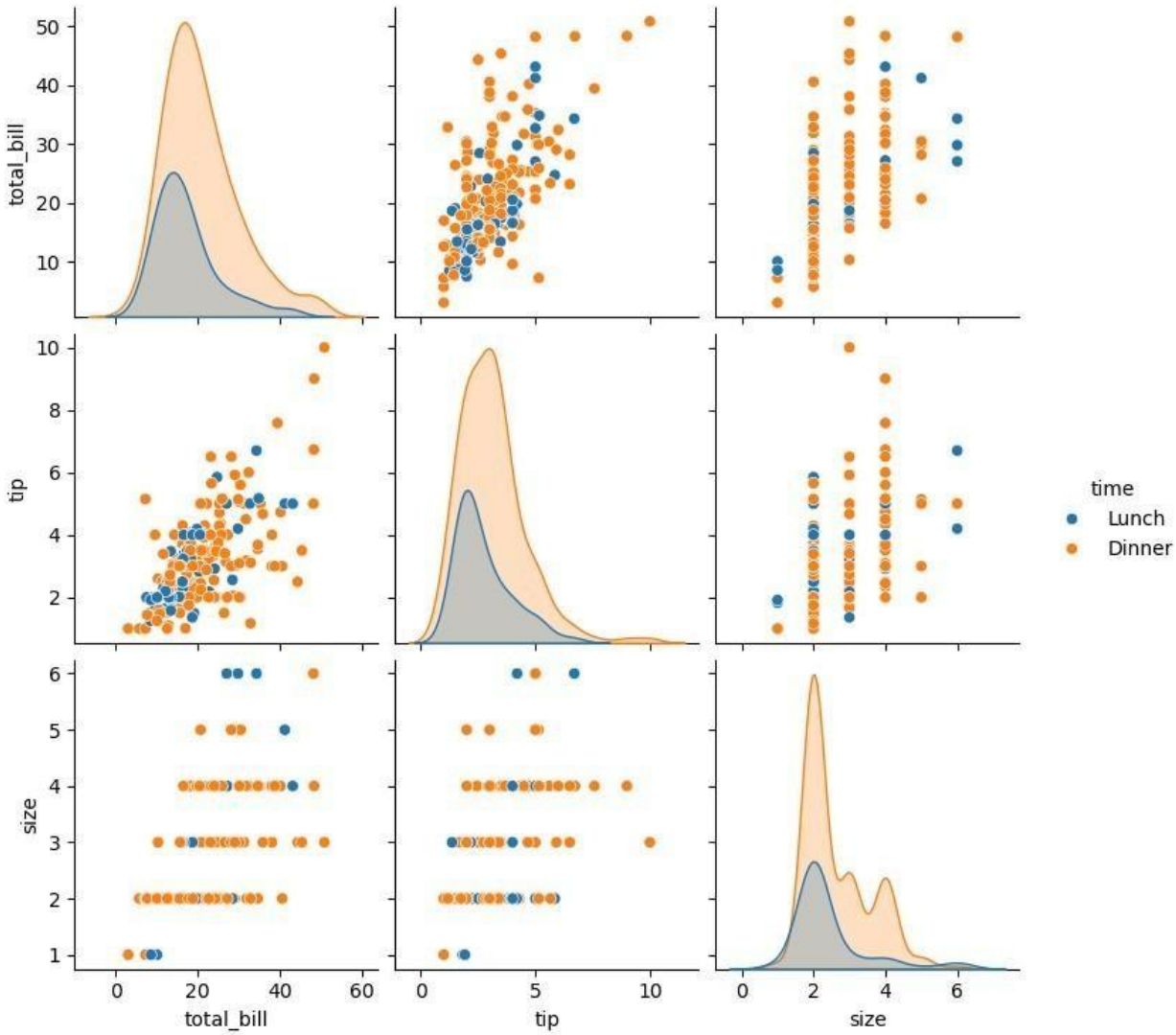

50
40
30
20
10
total_bill
2
4
6
8
10
tip

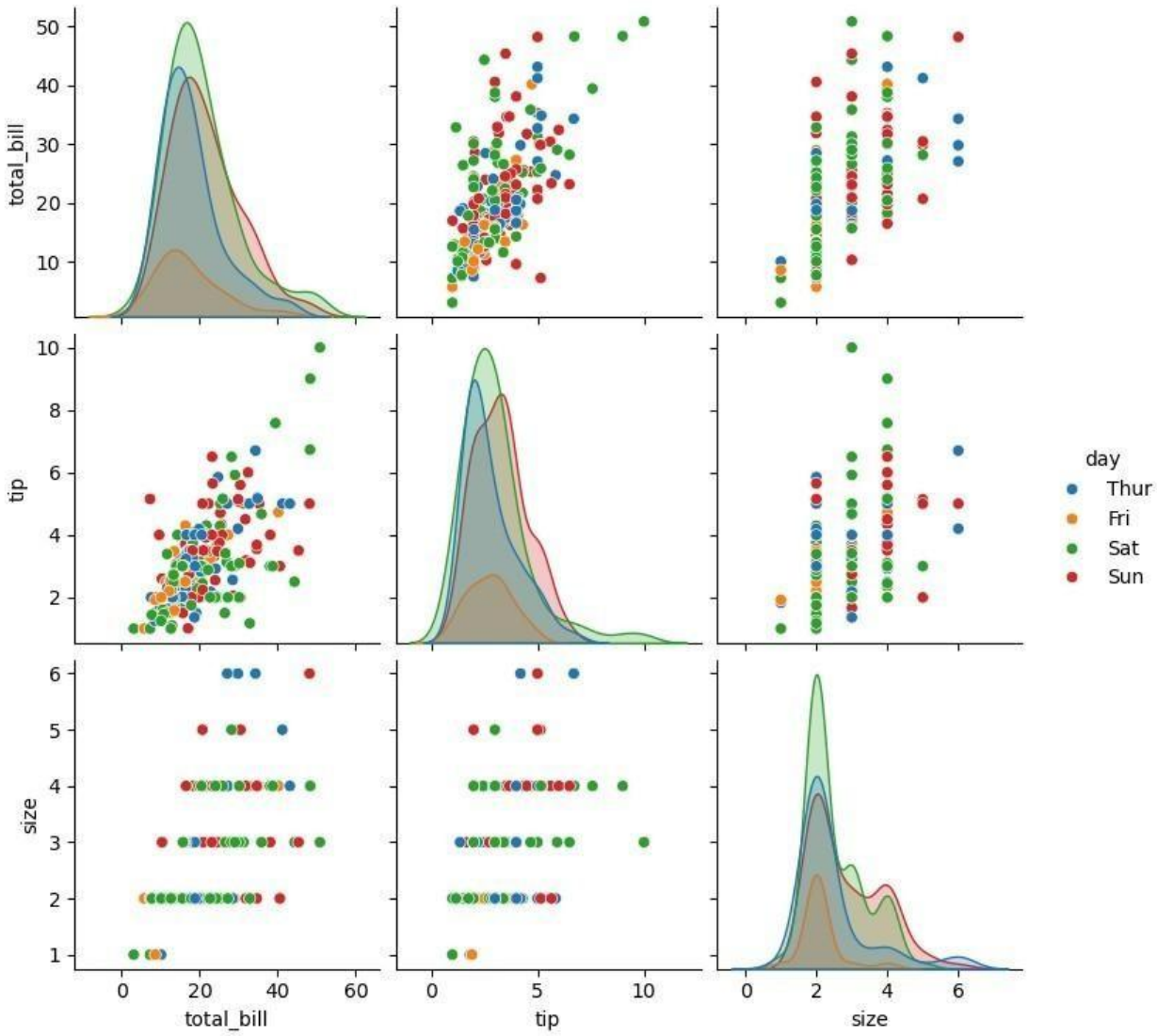


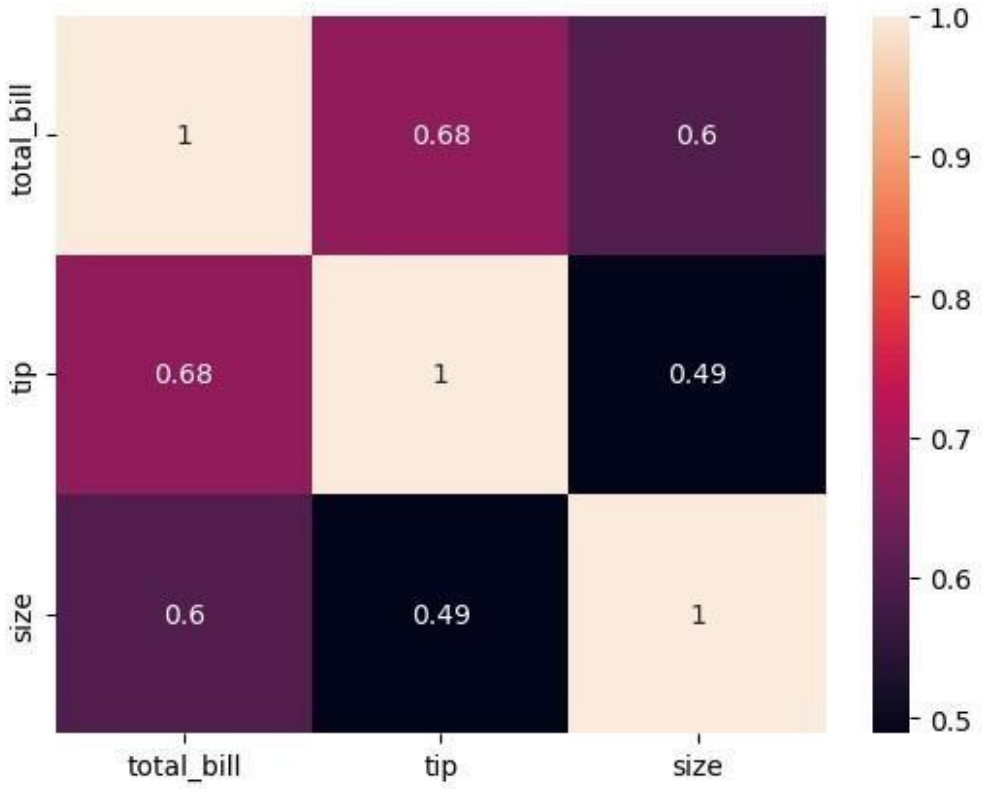

```
time

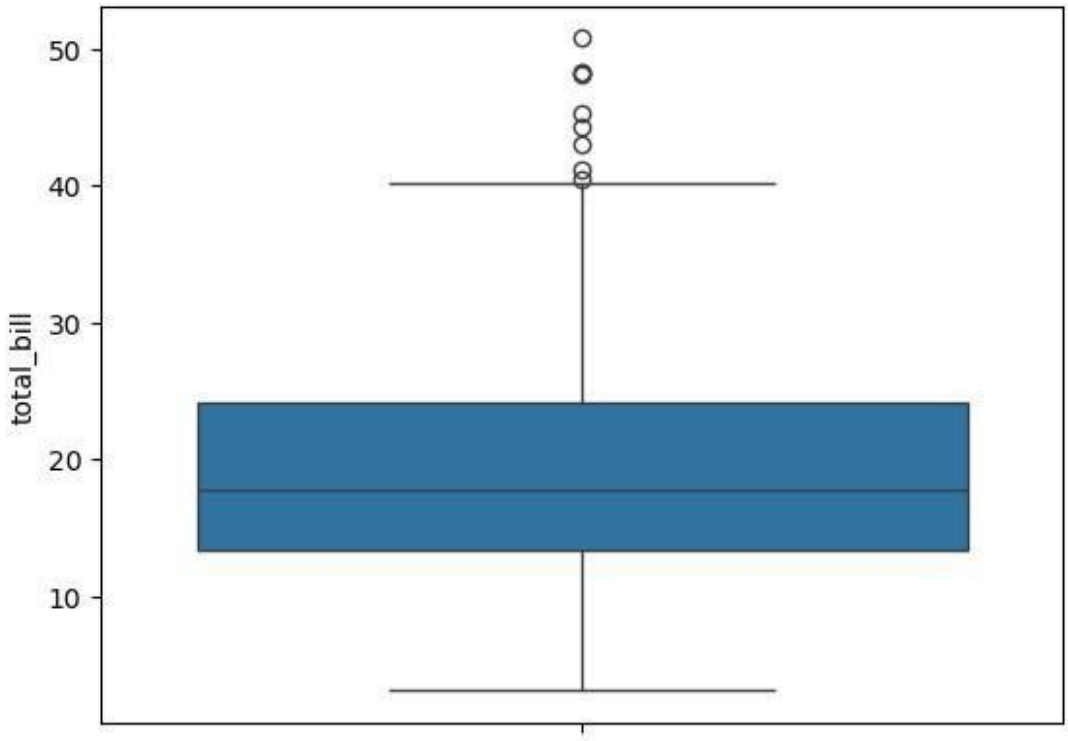
Dinner    176
Lunch      68

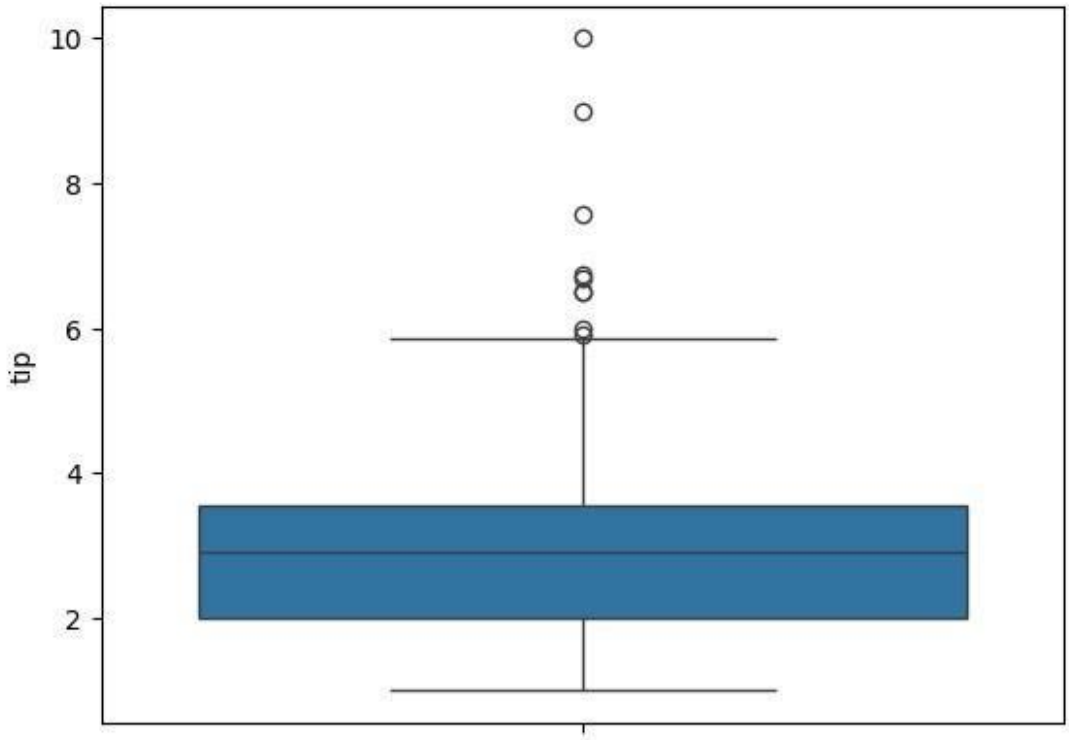
Name: count, dtype: int64
```

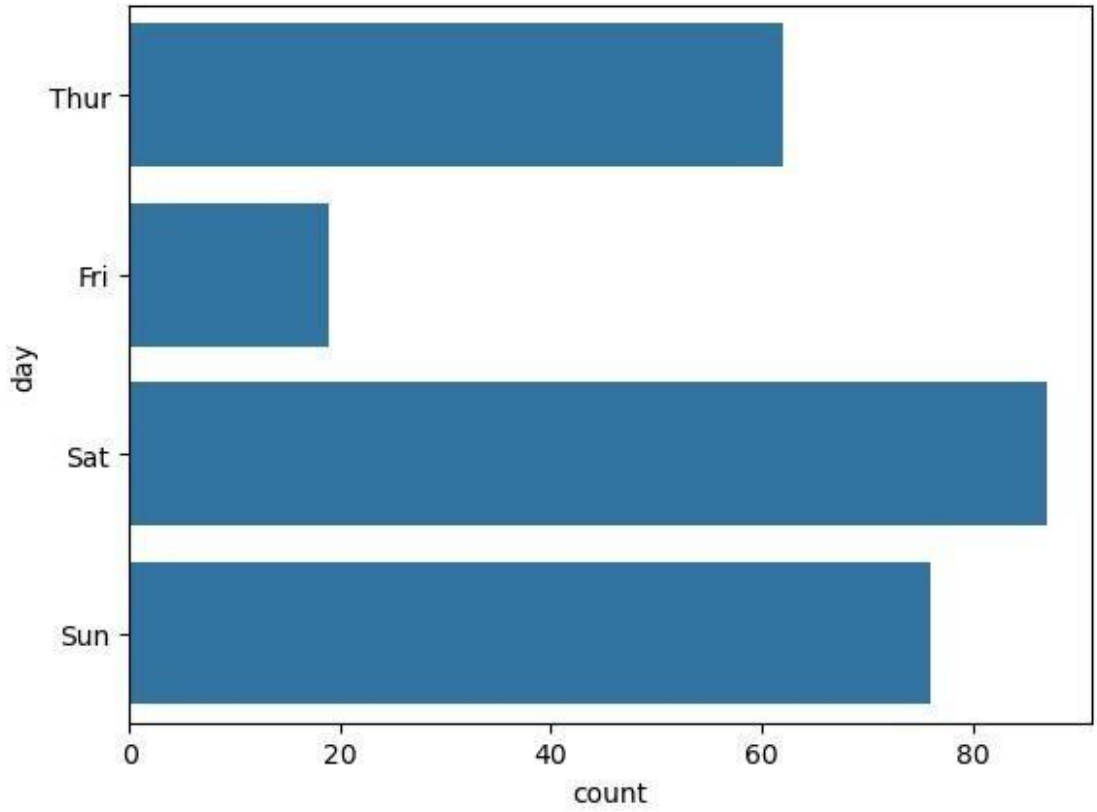

total_bill
tip
size
time
Lunch
Dinner

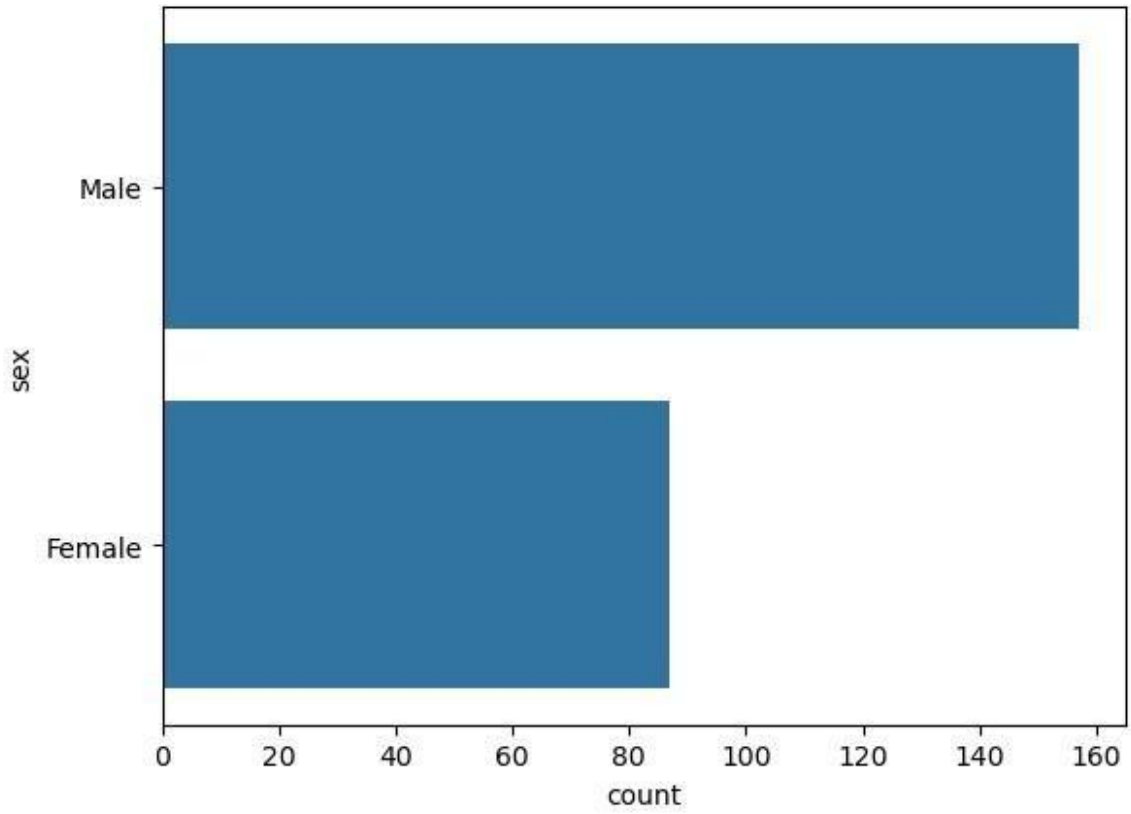

total_bill
tip
size
day
Thur
Fri
Sat
Sun

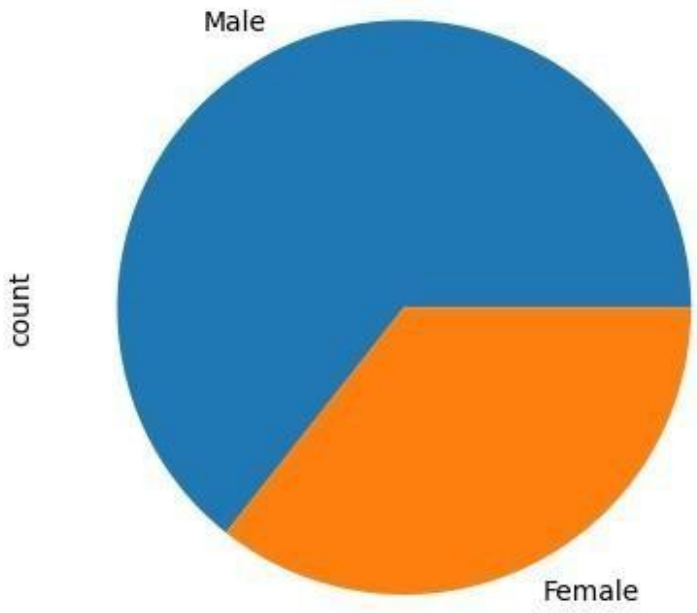

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

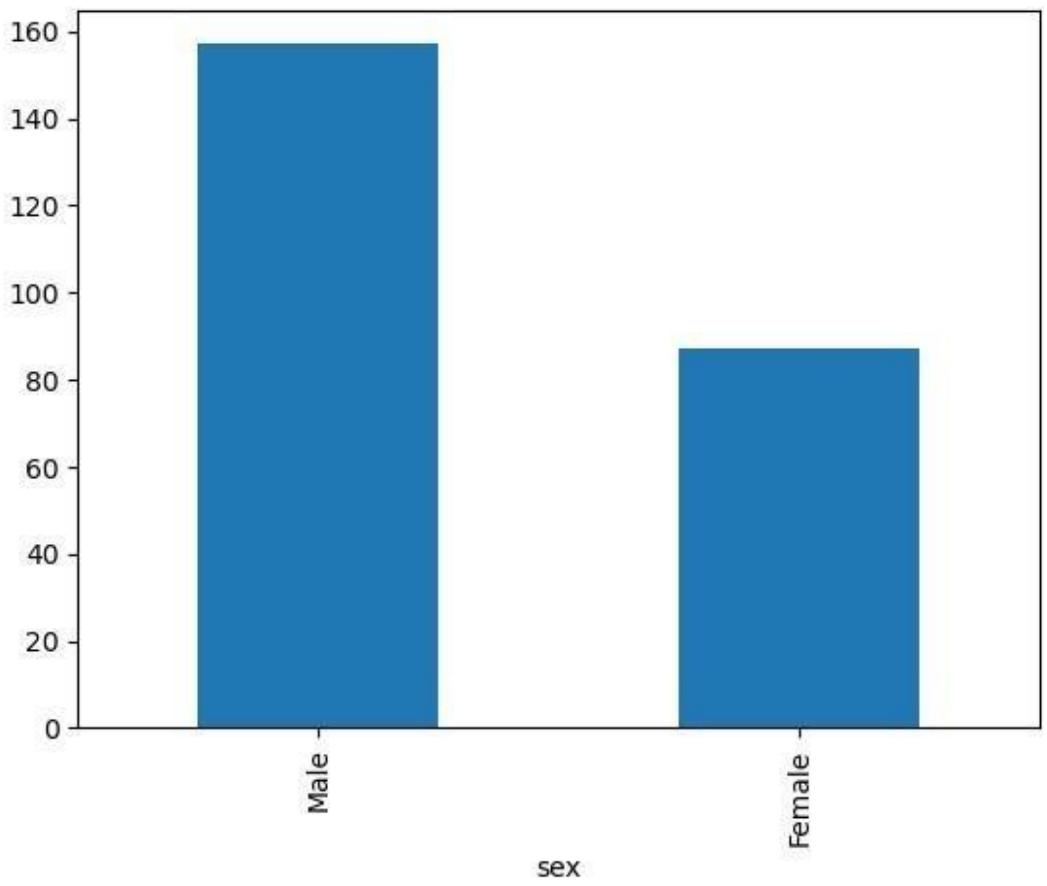

total_bill
50
40
30
20
10

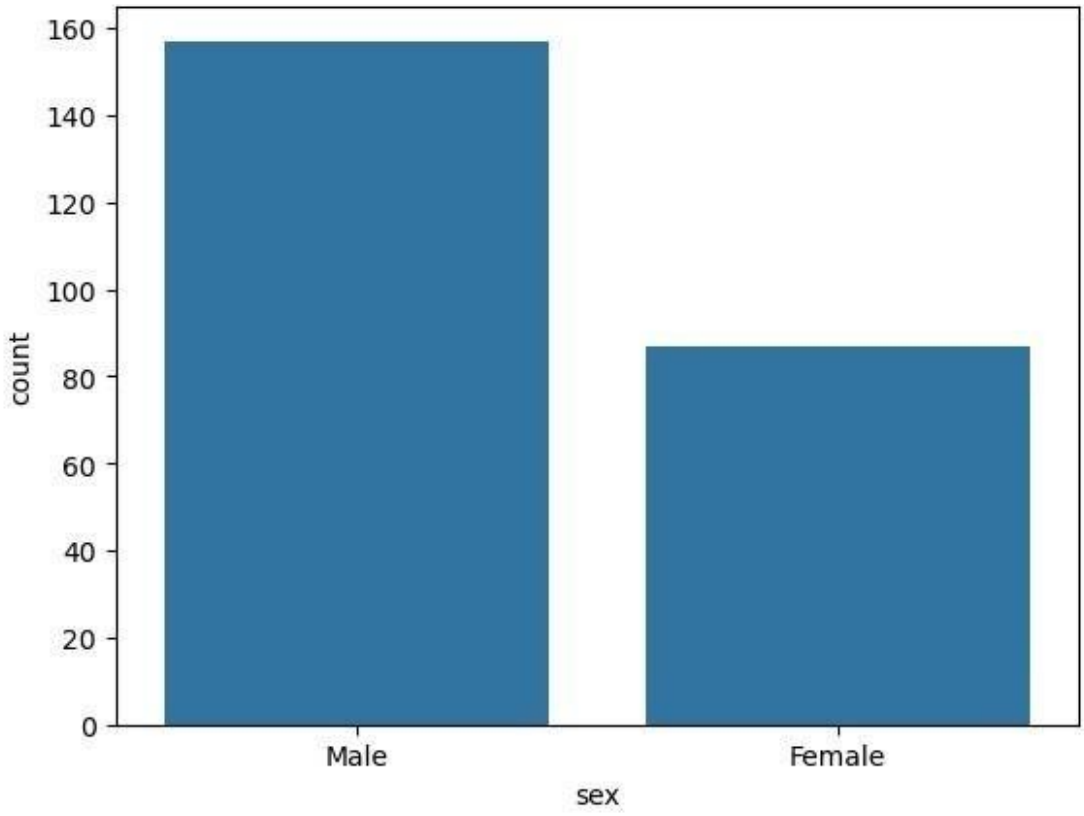

10
8
6
4
2
tip

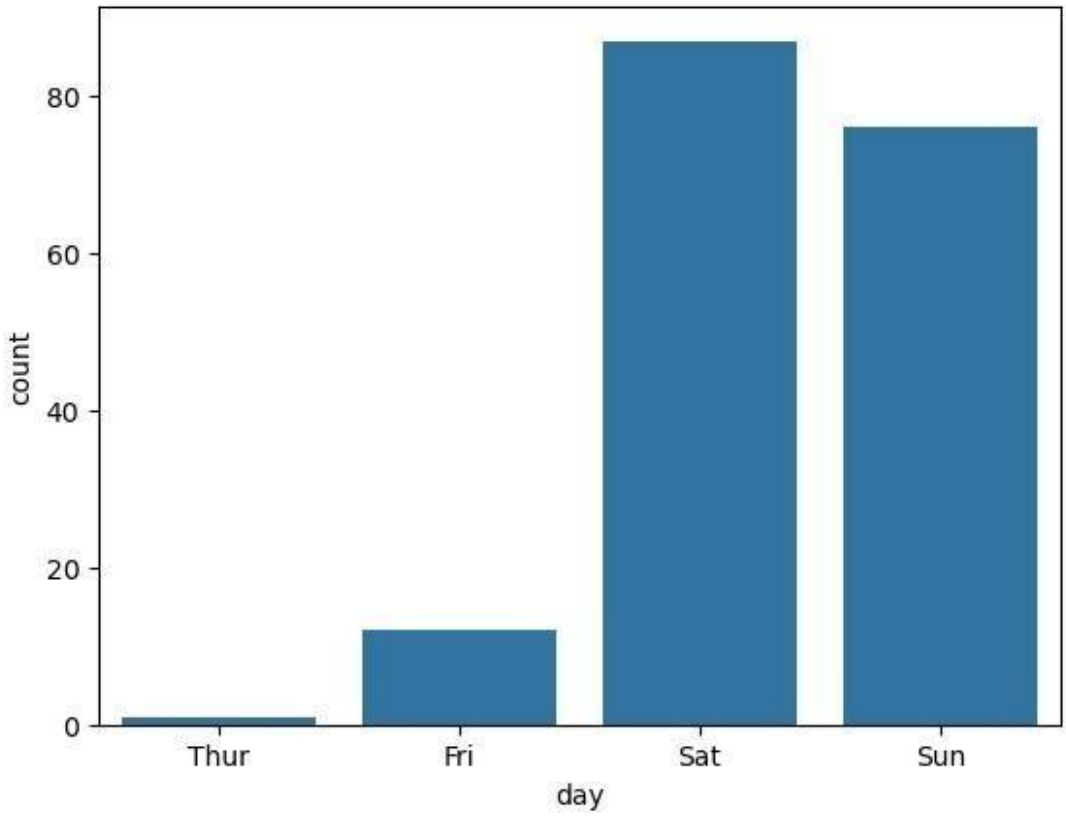

Thur
Fri
Sat
Sun
day
0
20
40
60
80
count


Male
Female
sex
0
20
40
60
80
100
120
140
160
count


Male
Female
count


160
140
120
100
80
60
40
20
0
Male
Female
sex


160
140
120
100
80
60
40
20
0
count
Male
Female
sex


80
60
40
20
0
count
Thur
Fri
Sat
Sun
day